Vedsai Vangapandu Computer Science Student | AI & Software Development

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Profile

Computer Science student at MacEwan University with hands-on experience in full-stack software development, AI integration, machine learning, and Python automation through research and technical projects. Skilled in Python, Java, C#/.NET, FastAPI, Spring Boot, React, Angular, PostgreSQL, and REST APIs. Experienced in developing AI-powered applications, designing secure backend systems, automating data workflows, implementing automated testing and CI/CD pipelines, and deploying containerized applications. Interested in AI engineering, enterprise software development, and building reliable, scalable solutions.

Education

Bachelor of Science - Computer Science Sep 2023 – Dec 2027
MacEwan University
Minors: Cybersecurity, Marketing
Relevant Coursework: Software Engineering, File & Database Management, Data Structures & Algorithms, Practical Programming Methodology, Human-Computer Interaction

Skills

- **Programming & Frameworks:** Python, Java, C#, .NET, TypeScript, JavaScript, SQL, FastAPI, Spring Boot, React, Angular, REST APIs
- **AI, Machine Learning & Data:** Gemini API, AI Agents, Generative AI, NLP, TF-IDF, K-Means Clustering, Regression, Pandas, NumPy, OpenPyXL
- **Backend, Databases & Cloud:** PostgreSQL, SQLAlchemy, Relational Database Design, Spring Security, Docker, Render, Neon, Firebase
- **Testing & Software Engineering:** pytest, JUnit, Testcontainers, Selenium, Cucumber, Playwright, Vitest, Git/GitHub, GitHub Actions, CI/CD, Agile/Scrum, Unit & Integration Testing
- **Analytics & Design:** Power BI, Tableau, Plotly, Excel, Figma, Technical Documentation

Projects

Offshoot - AI-Assisted Legacy Modernization Platform
Technologies: Python, FastAPI, Angular, TypeScript, C#/.NET, COBOL, PostgreSQL, Gemini API, Docker, GitHub Actions
- Architected and directed the development of an AI-assisted COBOL-to-C# modernization platform featuring automated business-rule extraction, sandboxed code execution, human approval workflows, and secure migration pipelines.
- Implemented verification workflows comparing legacy COBOL and generated C# across 500 test cases per module, detecting 16 rounding mismatches and validating the corrected implementation with 100% equivalence.
- Integrated automated testing, mutation analysis, and CI/CD pipelines, with secure Docker environments and cloud deployment using Render and Neon.

IknowServiceAI – AI-Powered IT Service Desk
Technologies: Python, FastAPI, React, PostgreSQL, Gemini API, SQLAlchemy, Docker, pytest, GitHub Actions
- Designed and directed the development of a full-stack AI-powered IT service desk integrating automated request classification, knowledge retrieval, citation validation, and intelligent recommendations using Google Gemini.
- Developed AI governance workflows incorporating deterministic security policies, human-in-the-loop approvals, role-based access control, privacy safeguards, and append-only audit logging.
- Established automated testing and CI/CD pipelines with 876 backend and 242 frontend tests, containerized the application with Docker, and deployed it using Render and Neon.

ClaimWillow – Enterprise Insurance Claims Management Platform
Technologies: Java, Spring Boot, React, TypeScript, PostgreSQL, Spring Security, JUnit, Testcontainers, Selenium, Docker
- Directed the development of a full-stack insurance claims management platform using Java, Spring Boot, React, and PostgreSQL, featuring role-based workflows, claim tracking, and a seven-state approval lifecycle.
- Designed six explainable validation rules for coverage verification, duplicate detection, and claim completeness, supported by optimistic concurrency control and database-enforced audit trails.
- Established automated unit, integration, and acceptance testing using JUnit, Testcontainers, Selenium, and Cucumber, integrating GitHub Actions CI/CD and deploying with Docker, Render, and Neon.

Professional Experience

Research Assistant – World Bank Economic & Governance Data Research Feb 2026 – Present

MacEwan University

- Develop Python automation workflows integrating the World Bank Indicators REST API to retrieve, process, and validate large-scale economic and governance datasets.
- Automated the transformation of 7,700+ raw records into 5,350 analysis-ready observations across 161 economies and five years, reducing repetitive manual processing.
- Debug and optimize Python scripts, resolve API integration and data-quality issues, and maintain reusable validation workflows using Python, OpenPyXL, and Excel.

Research Assistant – Product Recall & Consumer Safety Research Nov 2025 – Present

MacEwan University

- Develop Python-based data processing and automation workflows using Pandas, OpenPyXL, and Excel to analyze 9,793 product recall records across 82 variables.
- Automated approximately 9,700 longitudinal calculations and developed four reusable analytical measures, improving processing efficiency and reproducibility.
- Test, debug, and validate Python scripts to resolve data-quality issues and generate analytical reports and visualizations using Power BI, Tableau, and Plotly.

Research Assistant – Organizational Purpose & Nonprofit Performance Research Apr 2026 – Aug 2026

MacEwan University

- Developed Python automation and data-processing workflows using Pandas, NumPy, OpenPyXL, and python-docx to analyze 170+ academic studies with approximately 40 variables per source.
- Applied machine learning and NLP techniques, including TF-IDF, K-Means clustering, and regression, while debugging scripts, validating outputs, and automating research reporting.

Research Assistant & Product Designer – SAAF HCI Research Sep 2025 – Jul 2026

MacEwan University

- Designed and refined interactive UI/UX prototypes in Figma, translating user requirements into application designs and evaluating eight persuasive design features.
- Conducted usability evaluations with five HCI experts, incorporating feedback on privacy, trust, and usability into design improvements; first-authored a research paper accepted for SeGAH 2026.

Student Tutor Jan 2026 – Present

kihêw waciston Indigenous Centre, MacEwan University

- Provide individualized support in technical and quantitative coursework, applying analytical problem-solving, troubleshooting, and clear communication to explain complex concepts.
- Collaborate with students and university staff, adapting explanations to individual needs while managing multiple priorities and providing constructive feedback.

Organizations

Vice President – MacEwan AI Alliance (MAIA) Sep 2026 – Present

MacEwan University

- Collaborate with the executive leadership team to support initiatives in artificial intelligence, machine learning, and emerging technologies, promoting technical learning, student engagement, and innovation.
- Contribute to strategic planning, cross-functional collaboration, and knowledge sharing, supporting an inclusive AI-focused student community.

Director of Events, MacEwan Data Analytics Club Oct 2025 – Present

MacEwan University

- Coordinate event planning, technical workshops, speaker sessions, and project activities, collaborating with club executives, students, and external participants to support successful event delivery.
- Apply Agile-inspired coordination, problem-solving, and organizational skills to manage schedules, technical setup, resources, and multiple priorities while addressing logistical challenges.
- Facilitate cross-functional collaboration, stakeholder communication, and knowledge sharing, supporting technical learning initiatives and continuous improvement within the student community.

Student Volunteer Jan 2026 – Present

MacEwan International Badminton Club

- Collaborate with club members and organizers to coordinate weekly activities, manage equipment and event logistics, and support an organized, inclusive student environment.
- Demonstrate teamwork, adaptability, problem-solving, and communication skills while supporting club operations, resolving logistical challenges, and engaging with students.